

FFT Signal Processing — Pre-Analysis Notes

Natural Frequency Identification | Vibration Analysis | Research Paper Standard

Course: Manufacturing Systems Engineering
Topic: Experimental Modal Analysis using FFT
Standard: Written to meet peer-reviewed research paper methodology requirements

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1. What is Natural Frequency?

Every solid object has a unique internal "signature" — a set of frequencies at which it **prefers to vibrate**. This is its **natural frequency**.

Step	What Happens
You hit the material with an impulse (hammer / drop)	Energy is injected at ALL frequencies simultaneously
The material filters that energy	It absorbs most frequencies, vibrates ONLY at its own natural frequencies
The accelerometer captures the response	The recorded signal contains the structural "signature"
FFT decodes the signal	The peaks in the frequency spectrum = natural frequencies

Key Insight: Finding the natural frequency is how we decode the **physical stiffness and mass distribution** of a material. This is the foundation of Experimental Modal Analysis

| (EMA).

Paper-Ready Sentence

| "The acrylic specimen was subjected to an impulse excitation. The accelerometer response was analyzed spectrally to identify the dominant natural frequencies of the structure."

2. Sampling Rate — Connecting Time to Data Points

Your `.csv` file has **no concept of "seconds"** — it only stores row numbers. The **sampling rate** (F_s) is the bridge between time and row numbers.

How F_s is Calculated in MATLAB

```
matlab

dt = time(98) - time(97); % time gap between two consecutive rows (seconds)
Fs = 1 / dt;              % samples per second (Hz)
```

What the Numbers Mean

Row Number	Time (seconds)	Formula
Row 1	0 sec	—
Row 20,000	1 sec	$1 \times 20,000$
Row 40,000	2 sec	$2 \times 20,000$
Row 200,000	10 sec	$10 \times 20,000$

| **Rule:** `Row number = Time (seconds) ×  $F_s$`

Paper-Ready Sentence

| "The signal was acquired at a sampling frequency of $F_s = 20,000$ Hz ($dt = 50 \mu s$), yielding 200,000 data points over a 10-second record."

3. Selecting a Time Window (Spike Interval)

After plotting all 10 seconds, you visually identify the **clear impact spike** and select only that region for FFT analysis.

Example window: $t_1 = 2.725 \text{ s}$ to $t_2 = 2.850 \text{ s}$

Step 1 — Find Start and End Row Numbers

$$\begin{aligned}\text{Start Row} &= t_1 \times F_s = 2.725 \times 20,000 = \text{Row } 54,500 \\ \text{End Row} &= t_2 \times F_s = 2.850 \times 20,000 = \text{Row } 57,000\end{aligned}$$

Step 2 — Find Window Duration (Δt)

$$\Delta t = t_2 - t_1 = 2.850 - 2.725 = 0.125 \text{ seconds}$$

Step 3 — Find Number of Data Points (N)

$$\begin{aligned}N &= \Delta t \times F_s \\ N &= 0.125 \text{ s} \times 20,000 \text{ samples/s} = 2,500 \text{ data points}\end{aligned}$$

Why Multiply by F_s ? (The Core Logic)

F_s has units of **samples per second**.

So: $\text{seconds} \times (\text{samples/second}) = \text{samples}$

The units cancel perfectly $\rightarrow$ you get a **count of rows** inside the window.

Think of it like: if a car travels at 60 km/h for 0.125 hours $\rightarrow 60 \times 0.125 = 7.5 \text{ km}$. Same unit cancellation.

⚠ Important Correction Note

If your window is $2.725 \text{ s} \rightarrow 2.850 \text{ s}$, then $\Delta t = \mathbf{0.125 \text{ s}}$ (NOT 0.075 s).

Always verify: $t_2 - t_1$ before computing N. Professors will check this arithmetic.

Paper-Ready Sentence

"A window of $N = 2,500$ data points, corresponding to a duration of $\Delta t = 0.125 \text{ s}$, was extracted for spectral analysis based on visual inspection of the time-domain response."

4. Full Symmetry vs Half Symmetry

A raw FFT on $N = 2,500$ points outputs exactly **2,500 frequency-domain values**.
But **half of those are physically meaningless**.

Full Symmetry (Raw FFT — What You Get)

Points 1 $\rightarrow 1,250$: Positive frequencies ($0 \text{ Hz} \rightarrow 10,000 \text{ Hz}$) $\leftarrow$ KEEP THIS

The second half is a mathematically **exact mirror** of the first half — it contains **zero new information**.

Why Do Negative Frequencies Exist?

The Fourier Transform uses complex exponentials:

$$e^{i\omega t} = \cos(\omega t) + i \cdot \sin(\omega t)$$

This math **naturally produces** both $(+\omega)$ (positive) and $(-\omega)$ (negative) solutions.

But a physical acrylic bar **cannot vibrate at $-4,000$ Hz**.

Negative frequency is a **mathematical artefact** — not physical reality.

Half Symmetry (What We Keep and Use)

Action	Detail
Keep	Points 1 → N/2 (positive frequencies: 0 Hz to $F_s/2$)
Discard	Points N/2+1 → N (negative frequencies)
Correct amplitudes	Multiply all kept values $\times 2$ (except DC at 0 Hz)

Why Multiply by 2?

When you discard half the spectrum, you lose half the total signal energy.

Multiplying by 2 **restores the correct physical amplitude scale (m/s^2)**.

The DC component at 0 Hz is **not doubled** because it has no mirror counterpart.

⚠ If you skip the $\times 2$ correction: your amplitude axis will show values at **half the true physical magnitude**. Reviewers will catch this immediately.

Nyquist Frequency

$$\text{Nyquist Frequency } (F_n) = F_s / 2 = 20,000 / 2 = 10,000 \text{ Hz}$$

This is the **maximum frequency you can detect** from your data. Any frequency above F_n cannot be resolved — this is the **Nyquist-Shannon Sampling Theorem**.

Paper-Ready Sentence

"Due to the conjugate symmetry of the Fourier transform for real-valued time-series signals,

the two-sided spectrum was truncated to a one-sided spectrum, retaining only the physically meaningful positive frequencies from 0 Hz to the Nyquist frequency ($F_s/2 = 10$ kHz). Amplitudes were scaled by a factor of 2 to preserve spectral energy, with the exception of the DC component at 0 Hz."

5. Symbol Cheat Sheet

Symbol	Meaning	Units
F_s	Sampling frequency — rows collected per second	Hz
dt	Time between consecutive samples = $1/F_s$	seconds
t_1	Start time of selected window	seconds
t_2	End time of selected window	seconds
Δt	Window duration = $t_2 - t_1$	seconds
N	Number of data points in window = $\Delta t \times F_s$	count
F_n	Nyquist frequency = $F_s/2$ (max detectable frequency)	Hz
$N/2$	Number of useful FFT output points after half-symmetry	count
f	Frequency axis of FFT output	Hz
$X(f)$	FFT output (complex values — amplitude + phase)	—

6. Pre-FFT Checklist

Work through these **in order** before writing the FFT section of your code:

- ☐ **Step 1** — Plot all 10 seconds. Visually confirm where the impact spike is.
- ☐ **Step 2** — Note t_1 and t_2 from the plot (your spike interval start and end).
- ☐ **Step 3** — Compute $\Delta t = t_2 - t_1$ and verify the subtraction manually.
- ☐ **Step 4** — Compute $N = \Delta t \times F_s$. Write this number explicitly in your paper.
- ☐ **Step 5** — Extract rows from $(t_1 \times F_s)$ to $(t_2 \times F_s)$ in MATLAB.
- ☐ **Step 6** — Run FFT on the extracted window $\rightarrow N$ complex output values.
- ☐ **Step 7** — Truncate to first $N/2$ points (half-symmetry).

- ☐ **Step 8** — Multiply amplitudes $\times 2$ (except index 1, the DC component).
 - ☐ **Step 9** — Build frequency axis from 0 Hz to $F_s/2$.
 - ☐ **Step 10** — Plot. Identify the dominant peak $\rightarrow$ **that peak is your natural frequency**.
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7. Paper-Ready Sentences (Copy-Paste)

These are written to meet peer-review methodology standards. Adapt variable values as needed.

Describing Data Acquisition

"The signal was acquired at a sampling frequency of $F_s = 20,000$ Hz ($dt = 50 \mu s$), yielding 200,000 data points over a 10-second record."

Describing Window Selection

"A window of $N = 2,500$ data points, corresponding to a duration of $\Delta t = 0.125$ s, was extracted for spectral analysis based on visual inspection of the time-domain response."

Describing Half-Symmetry Truncation

"Due to the conjugate symmetry of the Fourier transform for real-valued time-series signals, the two-sided spectrum was truncated to a one-sided spectrum, retaining only the physically meaningful positive frequencies from 0 Hz to the Nyquist frequency ($F_s/2 = 10$ kHz). Amplitudes were scaled by a factor of 2 to preserve spectral energy, with the exception of the DC component at 0 Hz."

Describing Natural Frequency Identification

"The dominant peak in the one-sided power spectrum was identified as the first natural frequency of the specimen. This frequency corresponds to the fundamental mode of vibration of the acrylic bar under free-free boundary conditions."

8. Current MATLAB Code Base

This is the working code so far (pre-FFT stage):

```
matlab
```

```
clear; close all; clc; tic;

% --- Load data ---
data = readtable("Acrylic_0.csv");      % read .csv file
data = table2array(data);               % convert table to numeric array
time = data(:, 1);                      % column 1 = time (seconds)
x     = data(:, 2);                     % column 2 = acceleration (m/s2)

% --- Plot full 10-second signal ---
plot(time, x)
xlabel('Time (s)')
ylabel('Acceleration (m/s2)')
title('Full time-domain signal - Acrylic specimen')

% --- Sampling frequency ---
dt = time(98) - time(97);               % time step between samples
Fs = 1 / dt;                           % sampling frequency (Hz)

toc;
```

Next Steps to Add (FFT section — in progress)

matlab

```

% --- Step 1: Define spike window ---
t1 = 2.725; % window start (seconds)
t2 = 2.850; % window end (seconds)

% --- Step 2: Convert to row indices ---
idx1 = round(t1 * Fs);
idx2 = round(t2 * Fs);

% --- Step 3: Extract window ---
x_win = x(idx1:idx2);
N = length(x_win); % number of data points

% --- Step 4: FFT ---
X = fft(x_win); % full FFT (N complex values)

% --- Step 5: Half symmetry (one-sided spectrum) ---
X_half = X(1:floor(N/2)+1); % keep first N/2 points
X_half = abs(X_half) / N; % normalize
X_half(2:end-1) = 2 * X_half(2:end-1); % multiply by 2 (except DC)

% --- Step 6: Frequency axis ---
f = Fs * (0:floor(N/2)) / N; % frequency vector (0 to Fs/2)

% --- Step 7: Plot spectrum ---
plot(f, X_half)
xlabel('Frequency (Hz)')
ylabel('Amplitude (m/s²)')
title('One-sided FFT spectrum - Acrylic specimen')

```

References / Key Concepts

- **Nyquist-Shannon Sampling Theorem** — a signal sampled at F_s can only resolve frequencies up to $F_s/2$
 - **Conjugate symmetry** — for any real-valued input signal, $X(-f) = X^*(f)$, making the negative-frequency half a redundant mirror
 - **Experimental Modal Analysis (EMA)** — field of structural dynamics that uses impulse testing + FFT to identify natural frequencies, damping ratios, and mode shapes
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